Blind Tape

A five-year stock & options trading challenge on real market data

FMKT 256 · Fall 2026 · AI Extra Credit · Prompt 1: Finance app (trading game)
Team: Team member 1, Team member 2, Team member 3

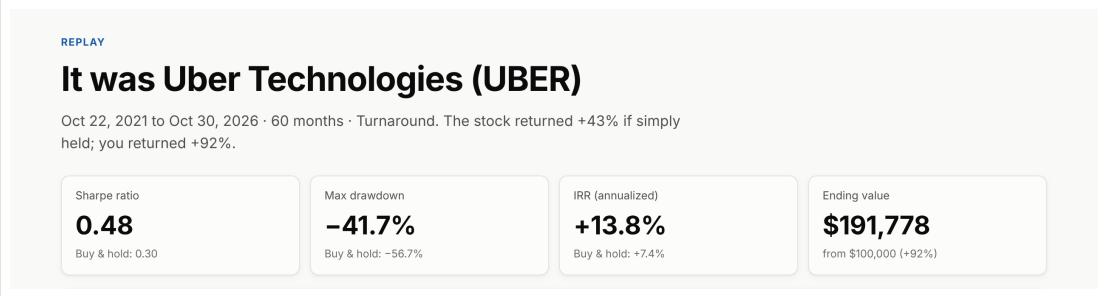
Play the game (no login, desktop or phone)	https://cartermburke.github.io/FMKT256-AI-Proj/
One-click demo game (replay)	Demo: a full 5-year blind game, scripted player
Source code, tests and data	https://github.com/cartermburke/FMKT256-AI-Proj
Daily data-update job (run history)	GitHub Actions: update-data
Live data status	Data page: series, sources, last update

What we built

Blind Tape is a browser trading game on real market data, built with Claude. Each player starts with a fixed **$100,000** and gets a stock drawn at random from 11 instruments. In blind mode the ticker and dates are hidden and the price is rebased to 100, so no one can trade on what they remember. The game steps through the **most recent five years** of daily prices one month at a time. Each month the player sets a long or short stock position and can trade one-month **calls and puts**. At the end the stock is revealed and the player is scored on **Sharpe ratio, maximum drawdown and IRR**, the criteria named in the prompt. The score is set against five autopilot strategies played on the same data with the same costs.

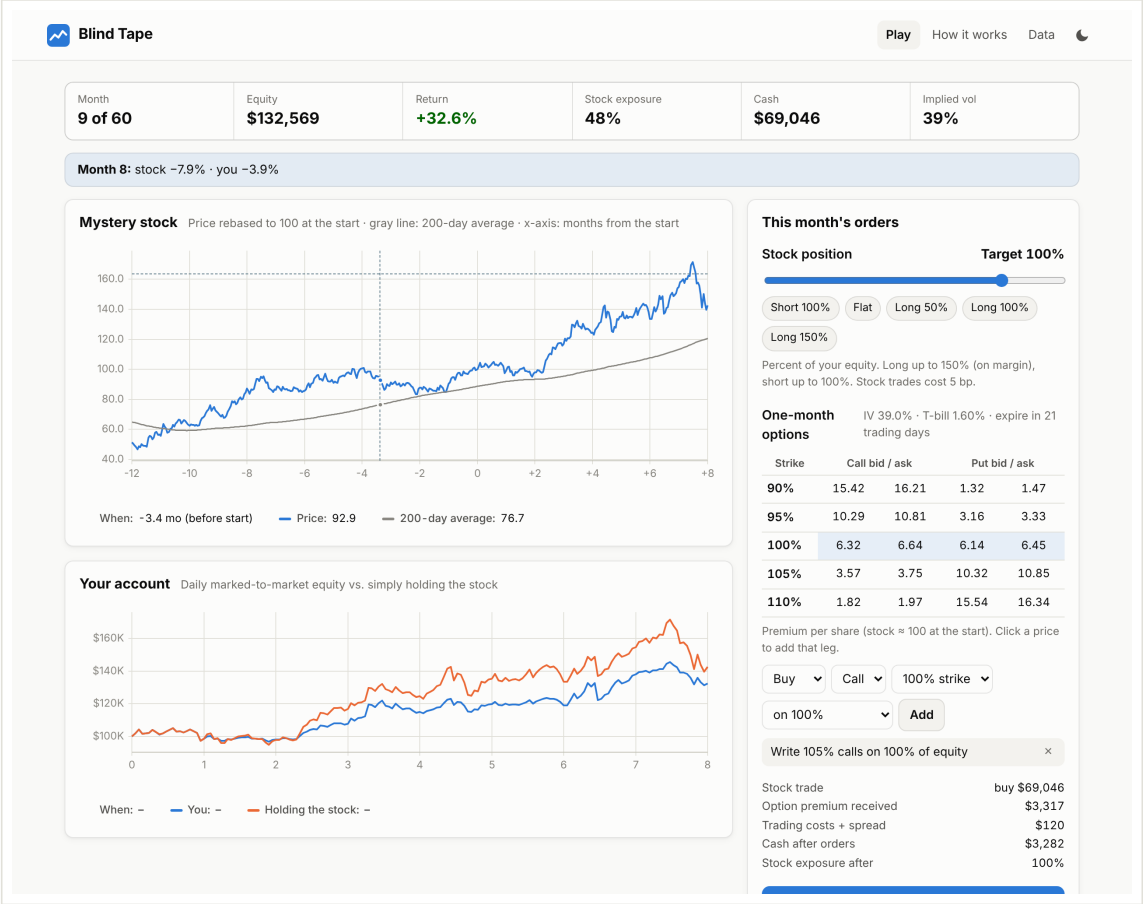
How it meets the prompt

- **Trading game:** fixed stake, one stock plus its derivatives over a fixed five-year period, judged on Sharpe, max drawdown and IRR (plus Sortino, volatility and beta).
- **Imports historical data:** about 7 years of daily adjusted closes for 11 instruments, the VIX and the 13-week T-bill yield, downloaded by code (yfinance with fallbacks) and validated before use.
- **Updates when new data arrive:** a scheduled GitHub Actions job runs every weekday after the close, appends the new bar, commits it and redeploys the site. The five-year window always ends at the latest close (currently October 30, 2026).
- **Repeatable without coding:** adding a stock to the game is one line in `config/tickers.json`; the next run downloads its history automatically.
- **No logins:** the game, the code and the data history are all public pages.



Playing a month

The left side shows the price history (up to a year before the start, plus every month played so far) with the 200-day average, and the account's daily marked-to-market equity next to simply holding the stock. The order ticket on the right sets the stock position and lists live option quotes at five strikes. Every order is checked before it can be submitted, and the ticket previews cash, costs and exposure.



Rules

- **Turns:** 21 trading days each; 60 turns for the full challenge, or a random 12-month slice in a quick round.
- **Stock:** −100% (short) to +150% (long on margin) of equity; 5 bp per trade.
- **Options:** one-month European calls and puts at 90, 95, 100, 105 and 110% strikes, cash-settled at expiry. You buy at the ask and sell at the bid.
- **Writing options:** written calls must be covered by shares; written puts must be cash-secured.
- **Financing:** cash earns the T-bill rate daily, margin debit pays T-bill + 1.5%, shorts pay 0.5% a year to borrow, and borrowing is capped at 50% of equity.
- **Risk controls:** margin is checked at every daily close. Equity below 25% of the gross stock position forces liquidation, and equity at zero ends the game.
- **Autopilot and replays:** an autopilot can finish the game with any benchmark rule, and every game produces a replay link that reproduces it exactly.

The scorecard and the debrief

At the end the stock is revealed. The demo game linked on page 1 is a scripted player on a blind five-year game. Its rule: long when the price is above its 200-day average and sell calls when the VIX is above 22; below the average, stay mostly in T-bills with put protection. The stock turned out to be **Uber Technologies (UBER)**. The scripted player had a Sharpe ratio of **0.48** vs 0.30 for buy & hold, a max drawdown of **42%** vs 57%, and an IRR of **13.8%** vs 7.4%. The same rule loses to buy & hold on most other stocks (next page), which is part of the lesson.

Scorecard Ranked by Sharpe ratio. Every strategy pays the same costs on the same data.

Strategy	Ending value	Total return	IRR	Sharpe	Sortino	Max DD	Volatility	Beta
Buy & hold the S&P 500 (SPY)	$178,253	+78.3%	+12.2%	0.54	0.77	−24.5%	17.1%	1.00
200-day trend (long 100% above the 200-day average, else cash)	$205,228	+105.2%	+15.4%	0.52	0.80	−43.0%	27.3%	0.52
Covered call (long 100%, write 105% calls monthly)	$207,637	+107.6%	+15.7%	0.52	0.73	−37.2%	29.3%	0.88
You	**$191,778**	**+91.8%**	**+13.8%**	**0.48**	**0.72**	**−41.7%**	**26.9%**	**0.56**
Buy & hold the stock	$143,310	+43.3%	+7.4%	0.30	0.45	−56.7%	44.8%	1.33
T-bills only	$120,958	+21.0%	+3.9%	0.00	0.00	0.0%	0.1%	0.00
Protective put (long 100%, buy 95% puts monthly)	$76,488	−23.5%	−5.2%	−0.11	−0.18	−49.3%	33.3%	0.82

Each row is ranked by Sharpe ratio. **Sharpe** is the mean daily return in excess of the T-bill rate divided by its standard deviation, × √252. **Max drawdown** is the largest peak-to-trough fall in daily equity. **IRR** is the XIRR of the account's cash flows. With a fixed stake that is the compound annual growth rate, and the engine computes a general XIRR so contributions could be added. A rule-based coach then explains the result: market timing, shorts, options P&L, frictions, and the comparison with the benchmarks.

✓ good **Market timing**

Across the 5 best months for the stock you averaged 65% exposure; across the 5 worst months, 45%. Your overall average was 58%.

You leaned in before rallies more than before sell-offs: real timing skill or luck, but it paid.

! watch **Options**

You traded options in 13 of 60 months. Net option P&L: -$3,668 (-0.7% of the starting stake per year). Bid/ask spread paid: $630.

Bought options lose their time value when the stock doesn't move far enough; insurance has a cost.

• note **Frictions**

Stock trading costs $633, borrow fees $0, net interest $9,477.

Interest is earned on idle cash at the 13-week T-bill rate and paid (T-bill + 1.5%) on margin debit.

✓ good **Versus the benchmarks**

Your Sharpe of 0.48 beat 3 of 6 benchmark strategies. Best benchmark: S&P 500 (Sharpe 0.54).

Simply buying and holding returned 43% with a 57% max drawdown; you returned 92% with 42%.

What five years of real data say about the benchmarks

Every game is benchmarked against five rules. Here they are on the full five-year window for every instrument in the pool (data through October 30, 2026). Cells show the Sharpe ratio; the best in each row is bold. The last two columns are for buy & hold.

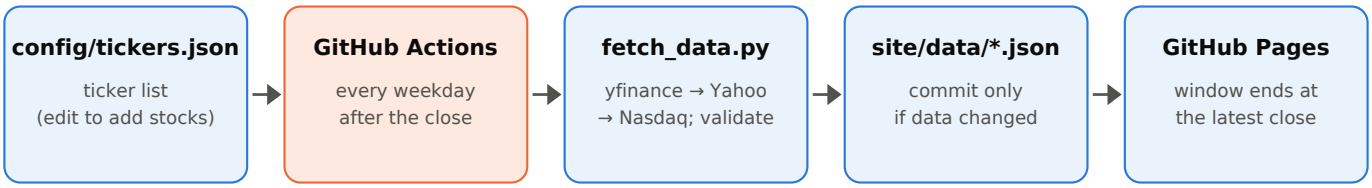
Stock	Regime	Buy & hold	Covered call	Protective put	200-day trend	S&P 500	B&H max DD	B&H IRR
NVDA	AI moonshot	**1.03**	0.98	0.90	0.99	0.54	-66%	+55.0%
INTC	Long decline	0.35	-0.07	0.46	0.50	**0.54**	-65%	+7.6%
PYPL	De-rating	-0.53	-0.39	-0.67	-0.19	**0.54**	-84%	-24.8%
MU	Memory cycle	1.13	0.90	0.89	**1.15**	0.54	-58%	+67.7%
TSLA	Wild swings	0.33	0.26	0.32	-0.33	**0.54**	-74%	+5.6%
KO	Low-vol defensive	0.68	**0.91**	0.40	0.47	0.54	-17%	+14.8%
XOM	Commodity boom	0.94	**1.03**	0.76	0.78	0.54	-21%	+28.6%
CRWD	High-vol growth	**0.67**	0.53	0.62	0.38	0.54	-68%	+28.6%
DIS	Sideways grind	-0.25	-0.42	-0.51	-0.70	**0.54**	-55%	-7.5%
UBER	Turnaround	0.30	0.52	-0.11	0.52	**0.54**	-57%	+7.4%
SPY	The market	0.54	**0.88**	0.32	0.35	0.54	-25%	+12.2%

Takeaways

- **Covered calls** beat buy & hold on Sharpe for 5 of 11 instruments. They win on low-volatility names and the index, where option premiums carry the most variance risk premium, and lag badly on the big trenders, whose upside they cap.
- **Protective puts** improved Sharpe for 1 of 11. Insurance helps in crashes but costs time value every month.
- **The 200-day trend rule** improved Sharpe for 4 of 11, usually by cutting drawdowns in long declines rather than by adding return.
- No single rule dominates, so a human who reads the tape well can beat all five. That is what makes the game worth playing.

How it works under the hood

Data pipeline: historical import and daily updates



If every source fails for a series, the last good data is kept and the site shows a stale-data banner.

Pricing options without historical option quotes

Free historical option chains for every stock don't exist, so options are priced with Black-Scholes on an implied volatility built from market data on each day:

$$IV^2 = RV_{63}(\text{stock})^2 + [\ (VIX / 100)^2 - RV_{63}(\text{S\&P 500})^2\ ]$$

RV_{63} is trailing 63-day realized volatility. The bracket is the market's variance risk premium that day: what investors pay for S&P 500 volatility beyond what the index is delivering. This prices SPY options exactly at the VIX and gives single stocks a modest premium over their own realized vol. **We changed this model during the build.** The first version multiplied each stock's realized vol by the index's VIX-to-realized ratio. Running the benchmarks on real data showed covered calls beating buy & hold on almost every stock (TSLA +197% vs +32%), which doesn't happen in real markets, because the volatility risk premium is concentrated in index options. The additive model fixed it.

Verification

- **23 unit tests** (0 failing) on the engine: Black-Scholes textbook values and put-call parity, the IV model, option settlement, covered and cash-secured rules, margin calls, short squeezes, interest accrual, and the Sharpe, Sortino, drawdown and XIRR math. They run on every push in GitHub Actions.
- **Independent cross-check:** a separate Python/pandas script recomputed 7 metrics on 66 real five-year equity curves (every benchmark on every stock): **0 mismatches** beyond 1e-6.
- **End-to-end browser test:** 28 of 28 checks pass. It plays games through the real UI in Chromium: valid and invalid orders, autopilot, the reveal, the replay link reproducing the exact result, the data page, dark mode and a phone-sized screen, with no console errors.
- **Adversarial review:** separate AI agents audited the finance math, tried to break the game, and played it as a first-time user. Their verified findings were fixed (next page).

How we used AI

- **Choosing the prompt:** Claude read the assignment and checked our real constraints: deadline, which data hosts were reachable, and existing tools. It then ran a six-agent panel. Three agents scoped one prompt each, and three judges scored the scopes as the grader, as a delivery lead and as a skeptical finance professor. The panel rated prompt 2 the safest. We chose prompt 1 because it was the most fun, and folded in the judges' warnings: test the math independently, never fake data, host on no-login pages, and fix the option-pricing bias.
- **Building:** Claude (Anthropic's Claude Code) wrote the game engine, the user interface, the data downloader and the GitHub Actions automation from a plan we reviewed and approved. We made the product decisions (game design, ticker pool, branding) and did the GitHub settings, play-testing and review.
- **Testing:** Claude wrote unit tests with hand-computed answers, an independent pandas recomputation of every metric on real data, and a browser test that plays the game through the UI. Running these caught real bugs during the build. One: the 5 bp trading cost made a maximum 150% long position fail the buying-power check. Another: chart axes lost their labels.
- **Adversarial review:** after the build, separate AI agents audited it independently. One checked the finance math and looked for look-ahead bias, one tried to exploit or crash the game, and one played it as a first-time student on desktop, phone and dark mode. A fourth looked for what the first three missed. Every finding had to come with a reproduction; we confirmed each one before fixing it.

Limitations

- Option prices are model prices, not historical quotes. There is no volatility skew, so out-of-the-money puts are probably cheaper here than they were in real markets.
- Decisions are monthly and fill at the daily close. Fractional shares are allowed, and there is no market impact beyond the fixed costs.
- Prices are dividend-adjusted (total return), so options are priced as if the stock paid no dividends.
- In blind mode a well-known stock can sometimes be recognised from the shape of its chart.

Run it yourself

Everything is in the public repository (https://github.com/cartermburke/FMKT256-AI-Proj). Open `site/index.html` through any static server; run `node --test tests/*.test.mjs` for the unit tests. To add a stock, edit `config/tickers.json` and push; the update job does the rest.

Academic exercise for FMKT 256, not investment advice. Results on historical data say nothing about future returns.